

Week 1 Task: Strategic Planning and Data Exploration in Logistics

Logistics Data Analyst Internship – Strategic Planning Report

Prepared by: Suhani Sallam

Project focus: E-commerce shipment and last-mile logistics analysis

1. Background and Logistics Scenario

This project proposes a data-driven approach for an e-commerce logistics operation that handles shipments between major Indian cities. The business objective is to improve delivery reliability, control transportation cost, and use vehicles and logistics resources more efficiently. The analysis focuses on shipment distance, weight, planned delivery time, actual delivery time, transport cost, fuel cost, vehicle utilization, tracking availability, and shipment damage.

The scenario is supported by the World Bank Logistics Performance Index (LPI), which evaluates logistics using dimensions such as customs efficiency, infrastructure, ease of arranging shipments, logistics-service quality, tracking and tracing, and timeliness. These dimensions show why both speed and reliability should be monitored rather than focusing only on cost.

2. Project Objectives

- Measure delivery performance using clearly defined KPIs.
- Identify routes and shipment profiles associated with delays or high cost.
- Understand the relationship between distance, shipment weight, and transport cost.
- Use descriptive analytics to support operational decisions.
- Demonstrate how regression and clustering could be used for logistics planning.
- Propose an end-to-end roadmap from data collection to predictive and optimization models.

3. Key Performance Indicators (KPIs)

KPI	Formula	Why it matters	Target direction
On-time delivery rate	$\text{On-time shipments} / \text{total shipments} \times 100$	Measures delivery reliability	Higher
Average delivery time	$\text{Total actual delivery days} / \text{shipments}$	Measures speed	Lower
Average shipment cost	$\text{Total transport cost} / \text{shipments}$	Measures cost efficiency	Lower
Vehicle utilization	$\text{Loaded weight} / \text{vehicle capacity} \times 100$	Shows resource usage	Balanced / higher
Damage rate	$\text{Damaged shipments} / \text{total shipments} \times 100$	Measures service quality	Lower

4.Data and Research Approach

The operational dataset included with this submission is synthetic demonstration data. It is intentionally labelled as synthetic and should not be presented as company data. The dataset contains 60 shipment records with fields for origin, destination, distance, planned and actual delivery time, shipment weight, transport cost, fuel cost, vehicle capacity, utilization, tracking availability, and damage status.

For external research, the World Bank LPI is used as a logistics benchmark. The World Bank states that the LPI covers 139 countries and measures the ability to move goods across borders with speed and reliability. The overall index combines six dimensions, including tracking and tracing and timeliness. This supports the decision to include delivery reliability and tracking-related metrics in the proposed operational dashboard.

5. Data Cleaning and Preparation

- a. Convert order dates to a proper datetime format.
- b. Check missing values and fill numeric missing weight values using the median.
- c. Fill missing tracking status with an explicit 'Unknown' category.
- d. Remove records with impossible values such as zero/negative distance or negative cost.
- e. Create derived fields such as delay days, on-time flag, and cost per kilometre.
- f. Check duplicate shipment IDs before production deployment.

6. Exploratory Data Analysis

The first analytical step is to compare KPIs overall and by route. A route-level groupby can calculate shipment count, average delivery days, on-time rate, and average transport cost. Routes with low on-time performance and high cost should be reviewed for operational causes such as long distance, low vehicle utilization, repeated delays, or poor tracking coverage.

Trend analysis can also show whether average transportation cost or delivery time changes over the period. Visualizations such as line charts, bar charts, and scatter plots can be used to make these patterns easier for managers to understand.

7. Data Science Methods

Regression can be used to estimate transport cost from measurable variables such as distance and shipment weight. The model can help estimate expected cost and identify unusual shipments whose actual cost is substantially higher than expected.

Clustering can group shipments into operational profiles, for example short/light shipments, medium-distance shipments, and long/heavy shipments. These groups can then be assigned different planning rules.

Optimization can be applied after the business constraints are defined. The objective could be to minimize total transportation cost while satisfying vehicle capacity, delivery-time, availability, and route constraints.

8. Strategic Roadmap

Phase 1 – Data collection: Collect shipment, route, vehicle, delivery, fuel, warehouse, and tracking data.

Phase 2 – Data quality: Validate IDs, timestamps, missing values, units, and duplicate records.

Phase 3 – Descriptive analytics: Build KPI tables and dashboards for route, city, vehicle, and time-period analysis.

Phase 4 – Diagnostic analytics: Investigate the reasons behind late and high-cost shipments.

Phase 5 – Predictive analytics: Build models to predict delivery time or transport cost.

Phase 6 – Optimization: Recommend routes/vehicle assignments subject to operational constraints.

Phase 7 – Monitoring: Track KPIs continuously and compare actual performance with targets.

9. Python Code Illustration

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.cluster import KMeans
df = pd.read_csv("logistics_shipments_demo.csv")
# Cleaning
df["weight_kg"] = df["weight_kg"].fillna(df["weight_kg"].median())
df["tracking_available"] = df["tracking_available"].fillna("Unknown")
# KPI features
df["delay_days"] = df["actual_days"] - df["planned_days"]
df["on_time"] = np.where(df["actual_days"] <= df["planned_days"], 1, 0)
df["cost_per_km"] = df["transport_cost_inr"] / df["distance_km"]
# Route-level summary
route_summary = df.groupby(["origin", "destination"]).agg(
    shipments=("shipment_id", "count"),
    avg_delivery_days=("actual_days", "mean"),
    on_time_rate=("on_time", "mean"),
    avg_cost=("transport_cost_inr", "mean")
).reset_index()
print(route_summary)
# Regression
X = df[["distance_km", "weight_kg"]]
y = df["transport_cost_inr"]
model = LinearRegression().fit(X, y)
# Clustering
features = df[["distance_km", "weight_kg",
"transport_cost_inr", "vehicle_utilization_pct"]]
df["cluster"] = KMeans(n_clusters=3, random_state=42, n_init=10).fit_predict(features)
```

10. Expected Outcomes

- a. A repeatable KPI framework for monitoring logistics performance.
- b. Identification of routes with weak on-time performance or unusually high cost.
- c. Better understanding of transportation cost drivers.
- d. Shipment segmentation that can support differentiated planning.
- e. A clear path from descriptive analytics to predictive modeling and optimization.

11. Conclusion

A logistics operation can benefit from combining operational KPIs with data science methods. The proposed workflow begins with reliable data collection and cleaning, moves to descriptive and diagnostic analysis, and then extends to regression, clustering, and optimization. The approach is designed to help managers make evidence-based decisions about routes, vehicles, delivery performance, and cost. The synthetic dataset demonstrates the technical workflow, while the World Bank LPI provides an external framework for understanding why reliability, tracking, infrastructure, service quality, and timeliness matter in logistics.

12. References

- f. World Bank. (2023). World Bank Releases Logistics Performance Index 2023. <https://www.worldbank.org/en/news/press-release/2023/04/21/world-bank-releases-logistics-performance-index-2023>
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